IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI VK TRADER

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

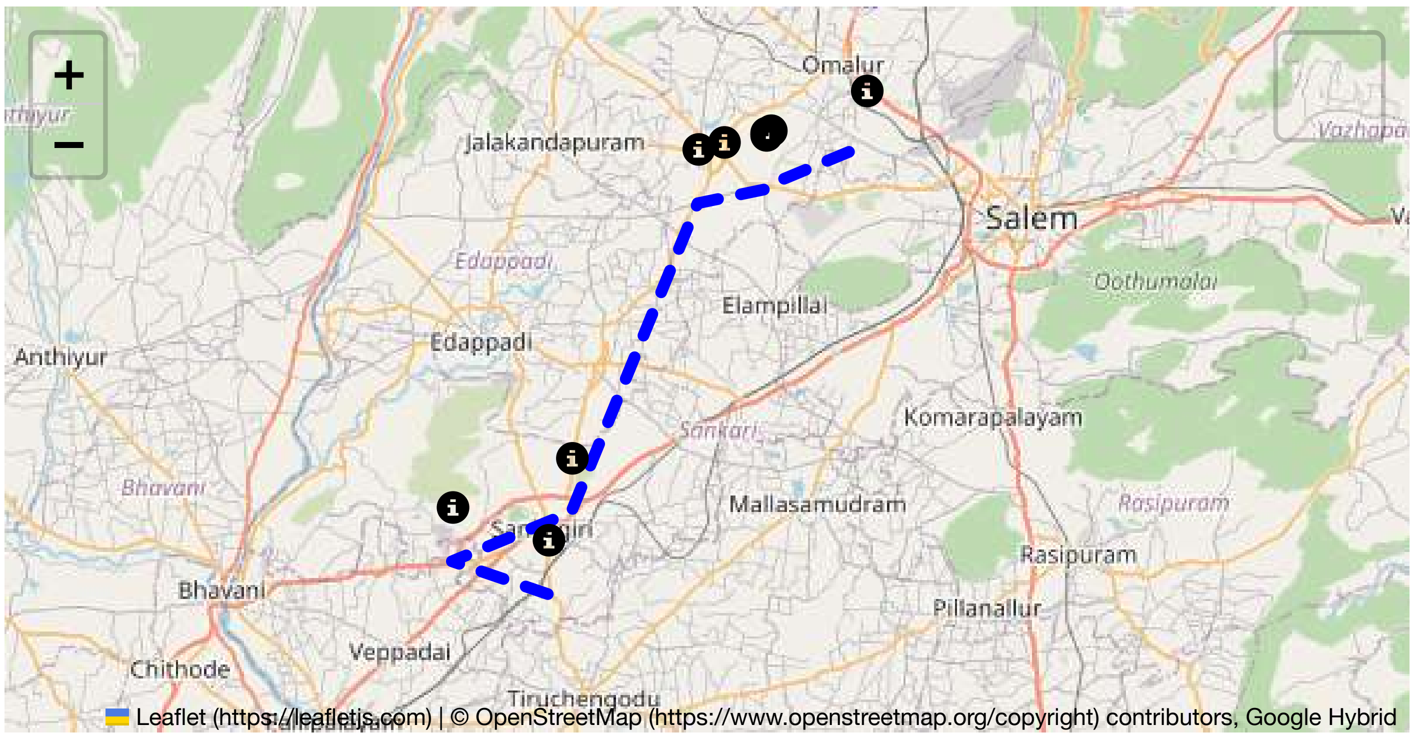
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

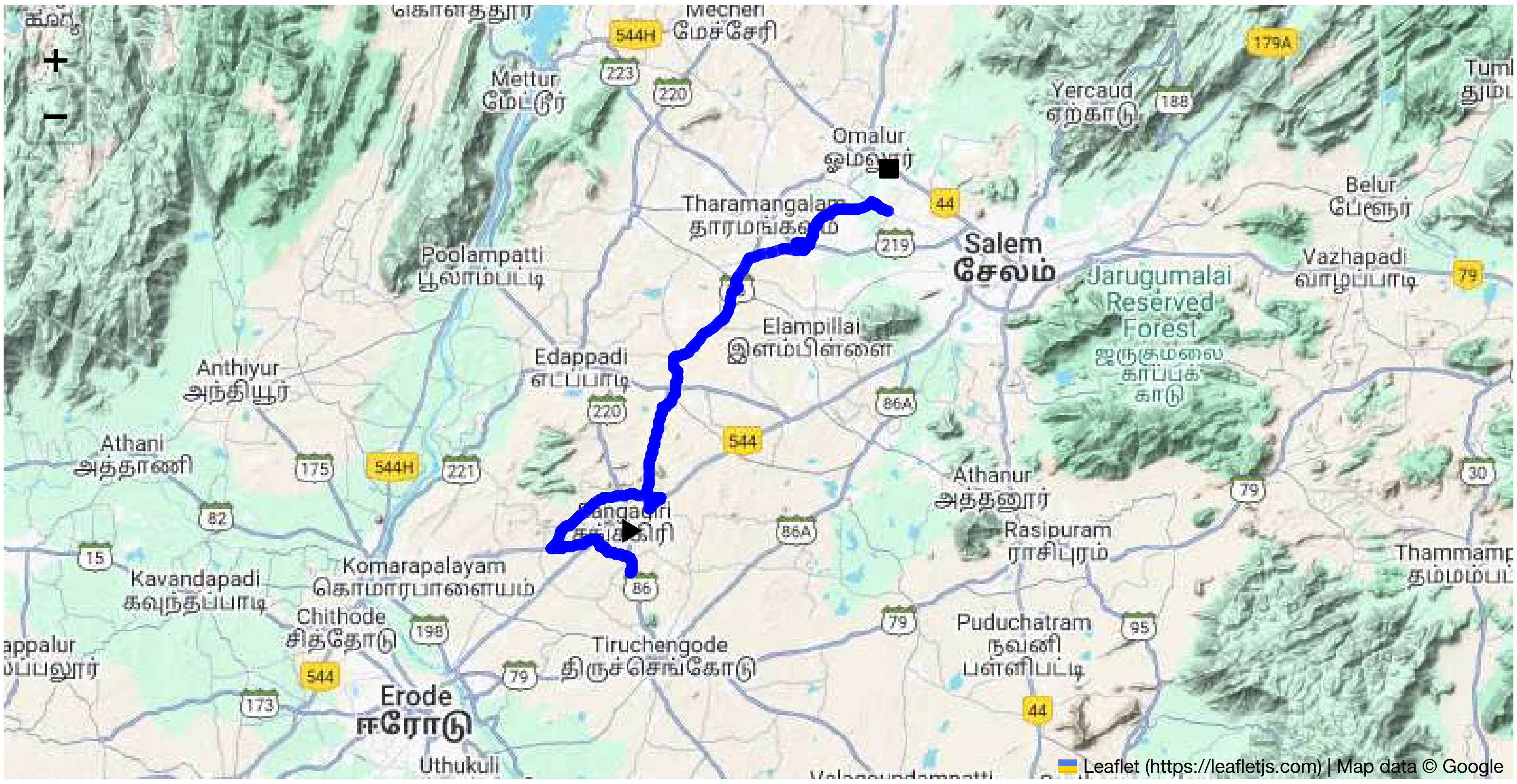
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 59.15 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours

Start: (11.4381, 77.8734)
End: (11.69658, 78.06129)



Welcome to the Journey Risk Management Study

Route Map Overview: The route covers approximately 59.15 kilometers, starting from CVQF+23W, Sangagiri, and ending at M3W6+MFG, Muthunaickenpatti Main Rd, Pagalpatti. It passes through major waypoints including FR48+7XQ Bhavani Main Road, FVPQ+96C SH 86, and various roads in Salem like Omalur and MX9G+5CR. The journey involves traveling mainly on state highways and main roads with variable traffic conditions and road quality.

Weather Conditions and Hazards: This region generally experiences a tropical climate. The hottest months are from March to June, with potential heavy rainfalls during the monsoon season (June to September). Visibility can be reduced during heavy rains, leading to increased risk of hydroplaning. It’s important to check weather forecasts during these seasons.

Traffic Patterns: The route encounters heavy traffic during peak hours, typically from 8 AM to 10 AM and 5 PM to 8 PM, especially in and around Salem City. Areas near Bhavani Main Road and Omalur are often congestion-prone due to market activity and local commuting.

Road Quality and Infrastructure: The quality ranges from good to moderate. State highways like SH 86 are generally well-maintained. However, roads inside Salem may have potholes or uneven patches. Roadworks and maintenance may occasionally cause disruptions, particularly near city intersections.

Alternative Routes: For emergencies, diversions via major state highways like SH 20 could be considered, providing alternate paths parallel to the main set route, especially bypassing densely populated areas in Salem.

Local Regulations: Tamil Nadu has specific regulations for transporting hazardous materials. Trucks should adhere to designated routes for hazardous material transport and are required to have appropriate signage. Checkpoints may require inspection of documentation and vehicle conformity to safety standards.

Historical Incidents: There have been incidents involving heavy vehicles in the past primarily due to driver fatigue and equipment failure, especially on the main roads intersecting in Salem City. Adherence to safety checks can mitigate these issues.

Environmental Considerations: Some segments pass through agricultural or semi-rural areas; care should be taken to avoid contamination. Compliance with environmental guidelines for spill prevention is crucial.

Communication and Connectivity: Mobile network coverage is generally good throughout the route except for potential dead zones on remote highway sections. Ensuring backup communication like satellite phones can be prudent.

Emergency Response Times: Emergency services in Salem City are relatively prompt, expected within 15-30 minutes. In more remote sections, response times could extend to approximately 45 minutes to an hour due to distance and road accessibility.

Overall Summary: The route from Sangagiri to Muthunaickenpatti entails moderate risk due to weather, traffic congestion, and road conditions. Adequate preparation through weather checks, optimized travel times to avoid peak hour congestion, and adherence to local regulations can significantly mitigate risks. Emergency protocols and communication readiness should be reinforced to ensure safety on the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44900, 77.87406	30 KM/Hr
4	Turn	Medium	11.45358, 77.85781	30 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	Medium	11.45839, 77.85143	30 KM/Hr
7	Turn	Medium	11.46053, 77.85034	30 KM/Hr
8	Turn	Medium	11.46307, 77.84941	30 KM/Hr
9	Turn	Medium	11.46316, 77.84921	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
12	Turn	High	11.49224, 77.89606	15 KM/Hr
13	Turn	High	11.49153, 77.89615	15 KM/Hr
14	Blind Spot	Blind Spot	11.48364, 77.88780	10 KM/Hr
15	Turn	Medium	11.49441, 77.88495	30 KM/Hr
16	Turn	Medium	11.49469, 77.88488	30 KM/Hr
17	Turn	High	11.55614, 77.89764	15 KM/Hr
18	Turn	High	11.58317, 77.90647	15 KM/Hr
19	Turn	Medium	11.63840, 77.94820	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	High	11.66252, 77.96048	15 KM/Hr
21	Turn	High	11.66821, 78.00077	15 KM/Hr
22	Blind Spot	Blind Spot	11.67303, 78.00187	10 KM/Hr
23	Blind Spot	Blind Spot	11.67370, 77.99409	10 KM/Hr
24	Turn	High	11.67340, 77.99404	15 KM/Hr
25	Turn	High	11.67327, 77.99475	15 KM/Hr
26	Turn	High	11.67356, 77.99482	15 KM/Hr
27	Turn	High	11.67276, 78.00377	15 KM/Hr
28	Turn	Medium	11.68005, 78.00701	30 KM/Hr
29	Turn	Medium	11.69847, 78.02406	30 KM/Hr
30	Turn	Medium	11.69784, 78.03990	30 KM/Hr
31	Turn	High	11.70268, 78.04958	15 KM/Hr
32	Turn	High	11.69948, 78.05515	15 KM/Hr
33	Turn	Medium	11.69868, 78.05541	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	Government Hospital, K.R.Thoppur	11.68382, 78.007937	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45358, 77.85781

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45353, 77.85697

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45839, 77.85143

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46053, 77.85034

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46307, 77.84941

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46316, 77.84921

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89606

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49153, 77.89615

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.48364, 77.88780

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49441, 77.88495

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49469, 77.88488

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55614, 77.89764

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58317, 77.90647

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63840, 77.94820

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66252, 77.96048

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66821, 78.00077

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.67303, 78.00187

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.67370, 77.99409

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67340, 77.99404

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67327, 77.99475

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67356, 77.99482

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67276, 78.00377

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68005, 78.00701

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69847, 78.02406

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69784, 78.03990

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70268, 78.04958

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69948, 78.05515

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69868, 78.05541